English translation prepared with ChatGPT (OpenAI, GPT-6), 26 October 2026, for translation of the Bulgarian text into English and LaTeX formatting. The Bulgarian originals remain the reference; translation may contain errors.

Probability and Statistics, CS Year 3, 2020/2021 Test 2: solutions — *Your grade is 2 + the number of points. Time allowed: 3 hours. Good luck! Work on* $(\Omega, \mathcal{F}, \mathbb{P})$.

Problem 1. A tank is filled with $X \in [0,1]$ units at the start of the week, and Y units are consumed during the week. Their joint density is

$$f_{X,Y}(x,y) = 3x\mathbb{1}_{\{0\le y\le x\le 1\}} = \begin{cases} 3x, & 0 \le y \le x \le 1, \\ 0, & \text{otherwise.} \end{cases}$$

1. (0.25 points) Find the probability that at most 1/2 unit is added and more than 1/4 consumed.

   Solution.
   $$\mathbb{P}(X < 1/2, Y > 1/4) = \int_{1/4}^{1/2}\int_{1/4}^{x} 3x\,dy\,dx = \int_{1/4}^{1/2} 3x(x-1/4)\,dx = \frac{5}{128}.$$

2. (0.75 points) Find $\operatorname{Cov}(X,Y)$.

   Solution. For $x \in [0,1]$, $f_X(x) = \int_0^x 3x\,dy = 3x^2$, so $\mathbb{E}X = \int_0^1 x3x^2\,dx = 3/4$. Similarly, for $y \in [0,1]$,

   $$f_Y(y) = \int_y^1 3x\,dx = \frac{3}{2}(1-y^2), \qquad \mathbb{E}Y = \frac{3}{2}\int_0^1 y(1-y^2)\,dy = \frac{3}{8}.$$

   Finally,
   $$\mathbb{E}[XY] = \int_0^1\int_0^x xy3x\,dy\,dx = \int_0^1 \frac{3}{2}x^4\,dx = \frac{3}{10},$$
   so $\operatorname{Cov}(X,Y) = \mathbb{E}[XY] - \mathbb{E}X\mathbb{E}Y = 3/160$.

Problem 2. (1 point) Information displays at public transport stops contain 10×100 diodes. Their high-quality components have exponentially distributed working lifetimes with mean 10 years. When fewer than 75% work, the information is often unreadable and repair is required. What is the probability of repair being needed after three years?

Solution. Each lifetime is Exp(1/10), and its survival probability is

$$\mathbb{P}(\text{Exp}(1/10) > 3) = \int_3^\infty \frac{1}{10}e^{-x/10}\,dx = [-e^{-x/10}]_3^\infty = e^{-3/10} =: p.$$

The indicators $X_i = \mathbb{1}_{\{\text{diode } i \text{ works after three years}\}}$ are $\text{Ber}(p)$, with mean p and variance $p(1-p)$. The finite variance and relatively large $n = 1000$ allow the CLT approximation

$$\mathbb{P}\left(\sum_{i=1}^n X_i < 750\right) = \mathbb{P}\left(\frac{\sum X_i - np}{\sqrt{np(1-p)}} < \frac{750-np}{\sqrt{np(1-p)}}\right)$$
$$\approx \mathbb{P}(N(0,1) < 0.66) = \Phi(0.66) \approx 74.54\%.$$

Problem 3. Model returns of assets A, B, C as independent $N(3,2)$, $N(3,3)$ and $N(1,10)$ variables. You have five investment units.

1. (0.25 points) How should you allocate them to maximize expected profit?

   Solution. Every $xA + (5-x)B$, $x \in [0,5]$, has the maximal expectation 15.

2. (0.25 points) Among those allocations, which minimizes variance?

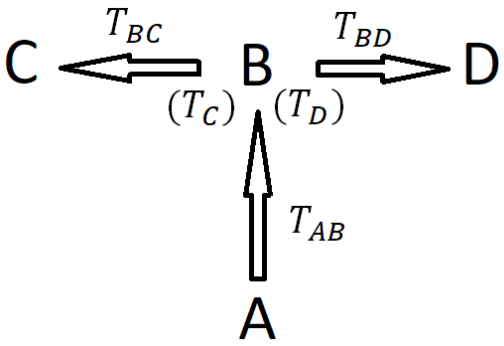
   Solution. $D(xA + (5-x)B) = 2x^2 + 3(5-x)^2 = 5x^2 - 30x + 75$, minimized at $x = 30/10 = 3$.

3. (0.5 points) A risk-taking investor puts five units into independent asset $D \sim N(-2, 20)$. What is the probability this outperforms part 2?

   Solution. The portfolio in part 2 is $N(15, 30)$, so

   $$\mathbb{P}(N(15,30) < N(-10,500)) = \mathbb{P}(N(25,530) < 0) = \Phi(-25/\sqrt{530}) \approx \Phi(-1.09) \approx 13.79\%.$$

Problem 4. Travellers X, Y go together from A to B, then wait for buses to C, D, respectively. The travel times are $T_{AB} \sim \text{Exp}(3)$, $T_{BC} \sim \text{Exp}(4)$ and $T_{BD} \sim \text{Exp}(5)$; the waiting times at B are $T_C \sim \text{Exp}(1)$ and $T_D \sim \text{Exp}(2)$. All these times are independent. Let ξ, η be the total journey times.



1. (0.25 points) Find $\mathbb{P}(T_C + \ln(\mathbb{E}T_D) > 0)$.

   Solution. As above, since $\ln(1/2) = -\ln 2$, the probability is $\mathbb{P}(T_C > \ln 2) = e^{-\ln 2} = 1/2$.

2. (0.75 points) Find $\mathrm{Cor}(\xi, \eta)$.

   Solution. $\xi = T_{AB} + T_C + T_{BC}$ and $\eta = T_{AB} + T_D + T_{BD}$. Independence gives

   $$D\xi = 1/9 + 1 + 1/16 = 169/144, \qquad D\eta = 1/9 + 1/4 + 1/25 = 361/900.$$

   Direct calculation shows

   $$\begin{aligned}\mathrm{Cov}(X_1 + X_2, X_3) &= \mathbb{E}[(X_1 + X_2)X_3] - \mathbb{E}[X_1 + X_2]\mathbb{E}[X_3] \\ &= \mathrm{Cov}(X_1, X_3) + \mathrm{Cov}(X_2, X_3).\end{aligned}$$

   Thus $\mathrm{Cov}(\xi, \eta) = \mathrm{Cov}(T_{AB}, T_{AB}) = DT_{AB} = 1/9$, yielding

   $$\mathrm{Cor}(\xi, \eta) = \frac{1/9}{\sqrt{D\xi D\eta}} = \frac{40}{247} \approx 0.1619.$$